

Data Ingestion

Data Example

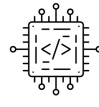
IMDb Movie Pathaan
★ Rating +1
👤 User Rahul



External Knowledge

"Give me complete details about 'Pathaan' - cast, music, plot, and other key information."

Temporal Hypergraph



Text Embedding

"Pathaan" text → → vector [0.23, -0.45, 0.67, ...]



KG Builder

Creates relationships:

- Pathaan -- starred_in --> Shah Rukh Khan
- Pathaan -- has_genre --> Action, Thriller
- Pathaan -- directed_by --> Siddharth Anand



Temporal Hypergraph

HYPEREDGE₁: {Pathaan, Shah Rukh Khan, Deepika Padukone, John Abraham} (single edge connecting movie + ALL 3 actors together)
Timestamp: 2023-01-25

HYPEREDGE₂: {Pathaan, Action, Thriller, Spy} (connects movie with ALL genres a once)

Enhanced Representation



Enhanced Community Detector

Identifies: "SRK Action Universe" community
Contains: {Pathaan, War, Tiger series, Don series}. Community embedding **captures** collective style.



Advanced Meta Path Reasoner

Finds reasoning paths:

- 1) User → liked War → starred SRK → also in Pathaan → Action genre
- 2) User → liked Tiger → similar director style → Pathaan **[Path score: 0.87 (high relevance)]**



Combined Embedding Generator

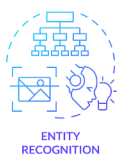
[hypergraph features] + [community features] + [meta-path features]

CONVERSATIONAL RECOMMENDATION



Users

"Mujhe kuch action dekhna hai" (I want to watch some action)



Extracts entities:

Genre: Action
Actor: None
Language: None
Mood: None



INTENT + RECOMMENDATION RECOGNIZER

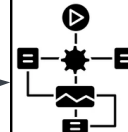
Intent: SEEK_RECOMMENDATION
Completeness: LOW (only genre known)
Decision: Need more information → trigger follow-up

Decides: recommend / ask / clarify



Graph Reasoner

Finds paths in hypergraph.



Recommendation Generator

Quantum scores + ranking



Follow up Question Generator

Checks missing info: [actor ×, language ×, style ×]

Generates strategic question:

"Which actors do you enjoy? Do you prefer high-intensity action or story-driven thrillers?"



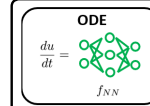
Response Generator

Natural language + explanation



: "Konse actors aapko pasand hain? Aur aapko zyada action chahiye ya thriller storyline?"

QUANTUM-INSPIRED BLOCK



Enhanced Neural ODE Block

Neural ODE: [0.8 action] --conversation--> [0.95 action] Continuous smooth evolution (not discrete updates)



Enhanced Quantum Block

Represents: "User is 60% action fan, 30% thriller fan, 10% uncertain". Allows exploring **MULTIPLE preference** hypotheses simultaneously.



Quantum Enhanced Hybrid GNN

Computes quantum score for "Pathaan":
score = $\text{Tr}(\rho_u \cdot P_{\text{Pathaan}})$ + entanglement_bonus(SRK, Action)
= 0.89 + 0.06 = **0.95 (very high!)**
For "Dilwale Dulhania Le Jayenge" (romance): score = 0.23 (low, as expected)